

TZU-YUAN HUANG (IAN)

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EDUCATION

Chair of information-oriented control, Technical University of Munich (TUM)

Ph.D. student, supervised by Prof. Sandra Hirche

Sep. 2023 - Current

Munich, Germany

- Research Direction: *Safe Generation of Robot Foundation Models*
- Project: *Robotics and Machine Learning in Automated Marine Litter Collection*

Dep. of Electr. Eng., National Cheng Kung University (NCKU)

Master of Science, supervised by Prof. Ming-Yang Cheng, overall GPA: 4.09/4.3

Sep. 2018 - Sep. 2020

Tainan, Taiwan

- Master's Thesis: *Study on Motion Planning for Industrial Manipulators Based on Kinematic Constraints and Dynamic Constraints* (Best Student Paper Award in Chinese Institute of Electrical Engineering (CIEE) and Robotics Society of Taiwan (RST))
- Best Presentation Paper, 2019 International Conference on Advanced Robotics and Intelligent Systems

Dep. of Mech. Eng., National Cheng Kung University (NCKU)

Bachelor of Science, overall GPA: 3.7/4.0

Sep. 2014 - Jun. 2018

Tainan, Taiwan

- Mechanical Engineering Capstone Project: A bionic teleoperated robot with Theo Jansen's leg mechanism and 4 Degree of Freedom (DoF) robotic arm
- Dean's list (top 10% academic performance of the class in the academic year, 2016)

EXPERIENCE

European Union's Horizon 2020 - SeaClear

Doctoral researcher

Sep. 2023 - Current

Munich, Germany

- Develop a torque-based grasping algorithm to lift marine litter up to 250kg with C++ and ROS2.
- Constructed Unity-based digital twin of Autonomous Underwater Vehicles in the marine environment

Syntec Technology Co., Ltd.

R&D Engineer

Nov. 2020 - Jul. 2023

Hsinchu, Taiwan

- Representative work: Real-time Spline-based trajectory smoothing algorithm in an embedded system
- Validation of unexpected disturbance torque detection and AI contour control function

Advance Technology Research Center, Delta Electronics, Inc.

Research Assistant

Jul. 2019 - Jul. 2020

Tainan, Taiwan

- Implemented a motion planning algorithm for smoothness optimization with NURBS curve

PEER REVIEWED JOURNAL PAPERS

1. **T.-Y. Huang**, S. Zhang, X. Dai, A. Capone, V. Todorovski, S. Sosnowski, S. Hirche, 2024, "Learning-based Prescribed-Time Safety for Control of Unknown Systems with Control Barrier Functions," *IEEE Control System Letters*.
2. **T.-Y. Huang**, M.-Y. Cheng, J.-L. Wong, 2025, "Quaternion-Based Velocity Scheduling for Robotic Systems," *Electronics*.
3. S. Zhang, D. Zhai, X. Dai, **T.-Y. Huang**, S. Hirche, 2025, "Gaussian Process-based Prescribed-Time Input-to-State Safety for Unknown System Control," *IEEE Transactions on Industrial Electronics*.
4. S. Zhang, X. Dai, D. Zhai, **T.-Y. Huang**, S. Hirche, 2025, "Event-triggered Online Learning for Safety-Critical Application with Control Barrier Functions," submitted to *IEEE Transactions on Automatic Control*.
5. H.-L. Huang, M.-Y. Cheng, **T.-Y. Huang**, 2023, "A Rapid Base Parameter Physical Feasibility Test Algorithm for Industrial Robot Manipulator Identification Using a Recurrent Neural Network," *IEEE Access*.

CONFERENCE PAPERS

1. **T.-Y. Huang**, A. Lederer, D.-J. Wu, X. Dai, S. Zhang, H.-C. Lin, S.-H. Sun, S. Sosnowski, S. Hirche, 2026, "SAD-Flower: Flow Matching for Safe, Admissible, and Dynamically Consistent Planning," *International Conference on Machine Learning (ICML)*
2. **T.-Y. Huang**, A. Lederer, N. Hoischen, J. Brüdigam, X. Xiao, S. Sosnowski, S. Hirche, 2025, "Toward Near-Globally Optimal Nonlinear Model Predictive Control via Diffusion Models," *Learning for Dynamics and Control System Conference (Selected as oral Presentation (8%))*

3. **T.-Y. Huang**, M.-Y. Cheng, B.-F. Wu, Y.-J. Lin, 2020, “Feedrate Planning for 6-DOF Industrial Manipulators,” International Conference on Advanced Robotics and Intelligent Systems.

4. Y. Origane, N. Hoischen, **T.-Y. Huang**, D. Kurabayashi, S. Sosnowski, S. Hirche, 2025, “Risk-Aware Trajectory Optimization and Control for an Underwater Suspended Robotic System,” *16th IFAC Conference on Control Applications in Marine Systems, Robotics, and Vehicles*

5. S. Zhang, DH. Zhai, X. Dai, **T.-Y. Huang**, Y. Xia, S. Hirche, 2024, “Learning-based Parameterized Barrier Function for Safety-Critical Control of Unknown Systems,” *63rd Conference on Decision and Control*.

6. J.-L. Wong, M.-Y. Cheng, **T.-Y. Huang**, JC Kuo, JM Wu, 2023, “On the Relationship between the Desired Feedrate and Contour Error Constraints for 2-DOF Robot Manipulators Performing Parametric Curve Following,” *International Automatic Control Conference*.

PATENTS

1. M.-Y. Cheng, **T.-Y. Huang**, J.-L. Wong, and C.-C. Hsu, 2022, “Method for planning feedrate of transmission mechanism based on physical constraints of joints”, Patent No. I766598, TW (Invention Patent).

ACADEMIC ACHIEVEMENTS AND AWARDS

- 2025 Oral presentation in Learning for Dynamics and Control System Conference, University of Michigan, USA
- 2025 EU-funded project SeaClear 2.0 is featured by CNN World
- 2025 Poster in RelAI annual report
- 2024 Presentation in 63rd Conference on Decision and Control, Milan, Italy
- 2020 3rd Place in Best Student Paper Award, Chinese Institute of Electrical Engineering (CIEE)
- 2020 Excellent Award in Master Thesis Award, Robotics Society of Taiwan (RST)
- 2020 Fellowship for research assistant in Dept. of E.E.
- 2019 Scholarship for outstanding academic performance, funded by Pan Wen Yuan Foundation
- 2019 Recruiting Fellowship of Delta Electronics, Inc.
- 2018 National Electric Vehicle Innovation Design and Construction Competition: 2nd in racing section, 3rd in technical report section
- 2018 Young College Elite of NCKU, selected 2/12000 for significant contribution and dedication to campus
- 2017 Undergraduate Research Fellowship (1 year), funded by Ministry of Science and Technology of Taiwan
- 2017 3rd Place in Creative Engineering Design Course final project competition

TEACHING EXPERIENCE

- TA in Adaptive Control (2024, 2025 summer), Advanced Control and Robotics Lab (2024, 2025 summer), Visual Servoing System (2019 Fall)
- Private C/C++ programming tutor
- Invited speaker in National Cheng Kung University Freshman Orientation(2015, 2017, 2018)

SUPERVISION

- Master’s thesis: “Near-Globally Optimal Approximation-based Model Predictive Control via Diffusion Models”
- Bachelor’s thesis: “Comparison of Generative-Model-based 6 DOF Robot Grasping”
- Master’s research project: “Optimal Grasping Manipulability with Constraint of Joint Position and Joint Torque”
- Bachelor’s research project: “Learning-based control for position tracking in 7 DOF robot manipulators”

LANGUAGES AND SKILLS

Language	Mandarin (Native), English (Fluent, IELTS: 7.0/9.0)
Software	C++, Python, Matlab, ROS 1/2, Mujoco, Pytorch
Skills	Generative model-based planning, Model predictive control, Control barrier functions

LEADERSHIPS & EXTRACURRICULAR ACTIVITIES

- CEO, Leadership Center, NCKU** Oct. 2017 - Jun. 2018
- Served as a school representative to advance the understanding, practice, and development of leadership for students
 - Collaborated with Acer Inc. to launch a two-day leadership camp, including 50+ participants
- Chairperson, Mechanical Engineering Technology Camp, NCKU** Jul. 2016 - Jun. 2017
- Led 60+ members to hold a 5-day camp, for 90+ senior high school students
 - Conducted industrial visit with Tongtai Machine & Tool Co., Ltd.